JIAER RUBBER & PLASTIC | 佳尔橡塑 · 高性能工业橡塑解决方案

MT51-WS-1 TECHNICAL DATASHEET

High-Performance Weather-Resistant & Water-Sealing Elastomer

Doc Ref: JE-TDS-MT51-WS-1 | Rev: 2026.10 | Grade: Industrial / OEM Certified

EXECUTIVE PRODUCT OVERVIEW & COMMERCIAL ADVANTAGE

JIAER MT51-WS-1 is an advanced, commercial-grade polymer & elastomer formulation engineered by Jiaer Rubber & Plastic (佳尔橡塑 · JIAERXIANGSU) for demanding industrial sealing, waterproofing, and insulation applications. Built to maximize operational uptime and lower lifecycle maintenance costs, MT51-WS-1 delivers superior tensile resilience and hydrophobic weather sealing.

- **Maximum ROI & Longevity:** Extended service life under continuous compression & UV.
- **Process & OEM Efficiency:** Consistent rheology for high-yield extrusion & molding.
- **Zero-Leakage Reliability:** Hydrophobic WS-1 barrier preventing moisture ingress.

TENSILE STRENGTH	ELONGATION AT BREAK	OPERATING TEMP	WATER RESISTANCE
⩾ 14.5 MPa	⩾ 420 %	-40°C to +125°C	WS-1 Grade
High Mechanical Integrity	Superior Elastic Recovery	All-Climate Thermal Stability	Hydrophobic Sealing Barrier

TECHNICAL SPECIFICATIONS & PHYSICAL PROPERTIES (MT51-WS-1)

*Standard Lab Conditions: 23°C ± 2°C, 50% RH

Property / Parameter	Test Standard	Unit	MT51-WS-1 Specification
Material Classification	JIAER / ISO 1629	—	Modified Weather-Sealing Elastomer (WS-1)
Appearance & Color	Visual Inspection	—	Smooth Homogeneous / Black or Custom OEM
Specific Gravity / Density	ASTM D297 / ISO 2781	g/cm³	1.18 ± 0.03
Hardness (Shore A)	ASTM D2240 / ISO 7619	Shore A	60 ± 5 (Customizable 50–75)
Tensile Strength (Min.)	ASTM D412 / ISO 37	MPa	⩾ 14.5
Elongation at Break (Min.)	ASTM D412 / ISO 37	%	⩾ 420
Tear Resistance (Die C)	ASTM D624	kN/m	⩾ 38.0
Compression Set (70°C × 22h)	ASTM D395 Method B	%	⩽ 22
Water Absorption (23°C × 24h)	ASTM D471	%	⩽ 0.8 (Hydrophobic WS-1 Grade)
Ozone & Weathering (72h)	ASTM D1149 (50 pphm)	Rating	No Cracks / Pass Grade A
Thermal Aging (100°C × 70h)	ASTM D573	Change	Hardness ⩽ +5 ShA | Tensile ⩽ -10%

HIGH-VALUE INDUSTRIAL APPLICATIONS

- Automotive & Commercial Weatherstripping
- Architectural Curtain Wall & Glazing Seals
- Waterproof Conduit, Cable & Pipe Gaskets
- HVAC Vibration Damping & Insulation
- Heavy Machinery Environmental Profiles

PROCESSING, PACKAGING & QUALITY

- Process: Extrusion, Compression & Injection
- Packaging: 25 kg barrier bags / 500 kg pallet
- Storage: 12 months in dry conditions (<30°C)
- Compliance: RoHS, REACH & ISO 9001 Certified
- Custom OEM: Hardness, color & cure profiles